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MILITARY SPECIFICATION

LOCOMOTIVES, DIESEL-ELECTRIC, 56-1/2-, 60-, 63-, or 66-INCH GAGE, 100-TON, 0-4-4-0 WHEEL FOR DOMESTIC OR FOREIGN SERVICE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 100-ton, single-engine, diesel-electric locomotive, mounted on two four-wheeled trucks for single-unit or multiple-unit operation in switching and road service on domestic or foreign railways.

1.2 Classification. The locomotive shall consist of one of the following gages, brakes, and couplers, as specified (see 6.2):

Gage
- 56-1/2 inch (143.5 cm) - domestic service
- 56-1/2 inch (143.5 cm) - foreign service
- 60 inch (152.4 cm) - foreign service
- 63 inch (160.0 cm) - foreign service
- 66 inch (167.6 cm) - foreign service

FSC 2210

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Brake equipment
Compressed-air
Vacuum

Coupler
Automatic AAR
Automatic Willison
Drawhook and screw

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-P-3321	- Painting and Marking: Railway Motive Power and Work Equipment.
MIL-P-3558	- Plates, Identification: Locomotives, Railway Cars, and Work Equipment.
MIL-L-58009	- Locomotives, Diesel-Electric and Associated Equipment: Inspection of.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1400	- Engines, Gasoline and Diesel, Methods of Test.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS35833	- Diagram, Equipment, Composite, Railway, 56-1/2, 60, 63, and 66 Inch Gages, Foreign Service.

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DRAWINGS

ME

13216E7927	- Broad Gage Fleet; End Arrangement of Coupler, AAR Type.
13216E7928	- Broad Gage Fleet; End Arrangement of Coupler, Drawhook and Screw Coupling.
13216E7929	- Broad Gage Fleet; End Arrangement of Coupler, Willison Type.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

INTERNATIONAL UNION OF RAILWAYS (UIC)

UIC Code No. 526-OR - Buffer Gear for Wagons.
UIC Code No. 520-OR - Drawgear.
UIC Code No. 548-OR - Automatic Vacuum Brake for Wagons Accepted for Running in Great Britain.

(Application for copies should be addressed to the International Union of Railways, General Secretariat, 14-16 Rue Jean Rey, Paris (15e), France.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Manual of Standards and Recommended Practices.
United States Safety Appliances for All Classes of Cars and Locomotives.
Electrical Manual of Standards and Recommended Practices.
Wheel and Axle Manual.

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Supplement to the Manual of Standards and Recommended Practices:

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

Standard No. 11 - Rotating Electric Machinary.
Standard No. 16 - Electric Railway Control Apparatus (ANSI C48.1).

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, 345 East 47th Street, New York, NY 10017.)

DEPARTMENT OF TRANSPORTATION (DOT) - FEDERAL RAILWAY ADMINISTRATION (FRA)

aws, Rules, and Instructions for Inspection and Testing of Locomotives Other Than Steam.

(Application for copies should be addressed to the U.S. Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The locomotive shall be 100-ton, 1000 horse power, 0-4-4-0 wheel, designed for foreign or domestic service, as specified, railway switching and road service in either single or multiple unit operation. Locomotive shall comply with the AAR Manual of Standards and Recommended Practice for locomotives of this type. The locomotive shall consist of an enclosed diesel-electric power plant, an electrical power transmission system, brake equipment that will permit the locomotive to move trains braked with pressure brakes or vacuum brakes, as specified, an operator's cab with control station, safety appliances and accessories, all mounted on an underframe with couplers, as specified, at each end. The running gear shall consist of two, two-axle motored trucks with an axle-mounted traction motor geared to each axle. Provisions shall be made in the design of the locomotive for holddown points and the lifting of the locomotive. The general design shall produce a locomotive with all parts accessible for maintenance and inspection.

3.1.1 Components. All electric components such as traction motor, traction generator of the locomotive shall conform to the requirements of IEEE Standard 11. All electric control apparatus shall conform to the requirements of IEEE Standard 16.

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3.2 First article.

3.2.1 First-produced locomotive. The contractor shall furnish a locomotive for examination and tests within the time frame specified (see 6.2), to prove that his production methods and choice of design will produce locomotives that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4, and shall be subject to surveillance and approval by the Government (see 6.5).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.10).

3.3.1 Bolts and rivets. Where aluminum construction is utilized, stainless steel or aluminum alloy containing 5 to 6 percent copper, or other AAR approved type of material, shall be used for bolts and rivets.

3.3.2 Substitution of materials. Materials not conforming to the specifications referenced herein shall not be substituted: (1) unless such substitutions are specifically approved by the contracting officer; (2) when it is clearly demonstrated that an improvement in operating characteristics, a saving in weight, conservation of critical or strategic materials, or a reduction in cost without sacrifice of reliability can be accomplished thereby; and (3) that such substitutions will not preclude the subsequent use of specified materials in effecting repairs or replacements. Requests for approval of such substitutions shall be accompanied by completely detailed supporting data which clearly illustrates the use of the proposed substitute material.

3.4 Threaded parts. Threaded parts shall conform to NBS Handbook H28.

3.5 Interchangeability. Jigs, templates, gages, and fixtures shall be used to insure interchangeability of like components and parts with respect to assembly, performance, and quality. All parts having the same part number shall be functionally and dimensionally interchangeable.

3.6 Safety. Safety appliances shall conform to AAR requirements for this type equipment. Electrical equipment shall be effectively guarded and grounded. All belts, pulleys, chains, gears, and other reciprocating or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel. All hot surfaces exposed to contact by personnel shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust, or electrical equipment. Steps, roof ladder, walkways, grab handles and other safety devices required, but not specifically stated, shall conform to AAR and FRA requirements.

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3.7 Serviceability.

3.7.1 Accessibility. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.7.2 Filling, draining, and checking provisions. Enclosures such as gear cases and oil sumps that contain a reservoir of lubricants for lubrication of enclosed parts shall be equipped with dipsticks, or sight gages to determine the lubricant level. Each enclosure shall be equipped for filling the enclosure with lubricants and for draining. The drain outlets shall be so located that removal of the plug will result in complete drainage of the lubricant from the enclosure. Integral tubes or troughs may be used to convey lubricant from the drain outlet to the ground. Drainage of battery compartment shall be located to prevent damage to other equipment. The drain outlets for the engine-cooling system shall provide complete drainage of the system. The radiator drain shall be such that coolant will not drain onto any parts of the locomotive. Extension hose is permitted.

3.7.3 Adjustment. All parts and components subject to adjustment or replacement by operating or maintenance personnel shall be accessible. Adjusting facilities where provided shall operate without interference and shall maintain such adjustments under all conditions of service.

3.7.4 Tolerances and fits. Tolerances and fits shall conform to all appropriate AAR manuals unless otherwise specified herein.

3.8 Environmental operation. The locomotive shall start and operate in ambient temperatures from minus 25° F (31.7° C) to plus 110° F (43.3° C) without additions. To provide safe operation and operator comfort, a cab heater and defroster shall be installed.

3.8.1 Engine heater. A heater shall be installed in each locomotive for operation at any ambient temperature from minus 25° F (minus 31.7° C) to plus 110° F (plus 43.3° C). The heater shall be a self-contained unit rated at not less than 50,000 British thermal units per hour (Btu's/hr) (14655 W). The heater shall provide heat for engine coolant, engine crankcase, lube oil, battery heating element, air compressor (and vacuum exhauster, when used) preferably in that order. Heater operation shall be entirely automatically controlled by temperature-actuated devices integral with heater (see 3.8.2). Heater shall operate satisfactorily on 64-74 volts direct current (dc) battery power, or 220 volts alternating current (ac), 60 Hertz, from an external power source. A 30-foot (9.4 m) heavy duty cord and plug, for 220-volt single phase, permanently connected

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to the heater, shall be provided. A bracket for the cable shall be provided adjacent to the heater. When operating on ac, heater equipment shall concurrently trickle charge the locomotive storage battery.

3.8.2 Temperature control. Temperature control equipment shall maintain temperature of system during standby with the engine shutdown to permit quick starting. Temperature control equipment, when operating under severe climatic conditions with engine idling, or lightly loaded, shall insure combustion of the fuel and lubrication of engine and related equipment. Control of automatic coolant radiator shutters (see 3.12.12.1), and other engine thermostats and controls, shall be integrated to insure specified performance of all equipment.

3.9 Dimensions and weights. Dimensions and weights of the locomotive shall be as follows:

Weight, fully loaded (locomotive)	200,000 pounds (90,800 kg) (approximately)
Weight, per driving axle	50,000 pounds (22,700 kg) (approximately)
Length, over end sills	40 feet 6 inches (12.3 m) (nominal)

Note. The difference in weight on the rails at each truck shall not exceed 2 percent of the total locomotive weight, fully loaded.

3.9.1 Clearances. The locomotive shall not exceed the dimensions shown on Plate B of AAR Manual of Standards and Recommended Practices or MS35833 as applicable.

3.10 Curvature. Locomotive alone shall be capable of negotiating a 193 foot (57.9 m) radius curve and with modifications negotiating a 90-foot (27.4 m) radius curve.

3.11 Capacity. Locomotive shall be capable of the following performance:

Maximum permissible safe speed (unloaded)	45 miles per hour (72.4 km/h)
Starting tractive effort at 25 percent adhesion, fully loaded	50,000 pounds (22,700 kg) (approximate)
Tractive effort - continuous rating at	30,000 pounds (19,100 kg) (approximate)

3.19 Design and construction. Design and construction shall be in accordance with the current-state-of-the-art. When specified (see 6.2), a structural analysis of the locomotive in accordance with the specifications stated above and indicating the design buff load shall be submitted by the contractor. The locomotive shall consist of the following components:

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3.12.1 Engine. For the purpose of selecting the engine, the locomotive shall be classified as Class I end item in accordance with MIL-STD-1410. The locomotive's power for propulsion shall be by a liquid cooled, multi-cylinder diesel engine equipped with all systems, components and accessories necessary for the required performance, and shall conform to MIL-STD-1410, Class I. The engine shall be of current commercial design that has proven acceptable for the type of service specified herein. The engine shall have sufficient rating at full load revolutions per minute (rpm) sufficient to provide a minimum of 1000 hp (745.7 kW) input to generator for traction based on 12-hour continuous duty at 8,000 foot (2438.4 m) altitude. Engine shall have a mechanical fuel injection system, recommended by the engine manufacturer, and which has proven acceptable in extensive commercial use and an electric motor-driven transfer pump for supplying fuel oil from tank to injection pump units. Engine shall idle smoothly for long periods of time and shall operate continuously on fuel oil conforming to DF-2, of VV-F-800 and on lubricating oil conforming to MIL-L-2104 at operating temperatures between minus 25° F (31.7° C) and plus 125° F (51.7° C). Cylinder liners shall be easily removable for replacement. The engine shall function as specified from sea level to approximately 8,000 feet (2438.4 m) altitude and capable of developing the power required while operating at standard barometric pressure (29.92 inches) (76.0 cm) with all auxiliary apparatus operating against maximum load, and without detonation, excessive vibration, or other deleterious effects. When under maximum load, smoke samples shall not indicate a reading above 5 percent at 86° F maximum air inlet temperature. An hour meter of not less than 9,999 hours for registering hours of engine operation shall be provided. Access to the end of the engine shaft or a shaft geared to the engine shaft for the use of a tachometer, shall be provided. The engine shall be governor-controlled over its entire throttle operating range. The engine exhaust pipe shall exhaust gases above the power plant and cab roof. An automatic engine overspeed safety device fitted with a positive manual reset mechanism shall be provided to protect the engine from overspeeding due to any cause.

3.12.1.1 Crankcase ventilation system. A positive crankcase ventilation system shall be provided on each engine. The system shall prevent or eliminate emission of crankcase vapors into the atmosphere under all operational conditions specified herein. If the ventilation system is connected to the air cleaner, it shall be connected in such a manner that no vapors or condensation will come in contact with, or be deposited on, the air cleaner element. The design of the system shall be such that service or replacement of the system, or any part thereof, may be accomplished without removal of other components of the engine.

3.12.1.2 Engine auxiliary equipment. All auxiliary equipment shall be readily accessible for inspection. The engine shall be furnished with all necessary auxiliary equipment, which shall include, but not be limited to the following:

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(a) Fuel filters.
(b) Oil filters.
(c) Excitor and auxilitary generator.
(d) Electric starting equipment (see 3.12.2).
(e) Engine cooling system.
(f) Radiator and cooling fan.

3.12.2 Generator, main. Generator shall be ruggedly constructed, shut wound, separately excited, railway type, direct-connected to, and supported on the inboard end by the engine or other approved construction in common use on commercial railroads. The armature shall have a single bearing and will be directly connected to engine crankshaft through flexible coupling. Generator shall have capacity to transmit to traction motors continuously the rated output of the engine. The starting winding shall be incorporated in the generator stator for battery starting of the diesel engine.

3.12.2.1 Auxiliary generator. The direct driven auxiliary generator shall provide current for control circuits, lighting, and battery charging and shall have an automatic voltage regulator.

3.12.3 Battery. Unless otherwise specified, a 64-volt, heavy-duty, lead acid type storage battery shall be provided for starting the engine and to furnish power for auxiliaries, controls and lights when engine is shut down. Battery shall be shipped in a fully charged condition unless otherwise specified.

3.12.3.1 Battery charging receptacles. Two battery charging receptacles shall be furnished, one on each side of the locomotive and as close to the battery compartment as is practical. Each receptacle shall be rated at 150 amperes and shall be in accordance with AAR Electrical Manual of Standards and Recommended Practices. A battery charging indicator shall be located at the operator's station.

3.12.3.2 Battery charging cables. A heavy duty cable 30 feet (9.4 m) in length shall be furnished for battery charging and engine starting. One end shall be fitted with a 150-ampere plug in accordance with AAR Electrical Manual of Standards and Recommended Practices. Means shall be provided for reeling and stowing the cable in the power plant enclosure.

3.12.3.3 Battery compartment. The battery compartment shall be ventilated to prevent accumulation of dangerous gases in cab and power plant compartment. Compartment shall be painted with acid-resistant paint. A battery switch shall be located in the control cabinet operated from the cab. Battery winter heating shall be by means of a noncorrosive heating panel or element, in each battery box, with radiating surface of not less than 8 square feet (2.5 m). Valves shall be provided to shut off heat when not required.

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3.12.4 Fuel system. The fuel system shall have a minimum capacity of 610 gallons (2271 L) with filler connections and fuel level gages furnished on each side of the locomotive. A remote reading indicator gage shall be located on the instrument panel at operator's position, suitable for 64-volt dc operation. Fuel tank will be equipped with splash plates, sediment bowls, vent pipes, cleaning holes, drain valves, and shall be arranged to prevent a fire hazard. Filling connections shall be inclined not less than 30 degrees, preferably 90 degrees, from the horizontal with sufficient heap for the fuel to flow freely into the tank. Filling connections shall be marked "FUEL" and the fuel cap shall be attached by a corrosion-resistant chain. Fuel lines shall be securely clamped to prevent vibration. Connections between the frame and the engine shall be by flexible tubing.

3.12.4.1 Emergency fuel shutoff. Three emergency fuel shutoffs shall be furnished. One shall be located in the operator's cab and one on each side of the locomotive.

3.12.5 Transmission. The locomotive transmission shall transmit the engine power to the wheels for either direction of travel, at any speed up to maximum (see 3.11). The transmission shall consist of a main generator directly connected to the engine, and through control apparatus, to traction motors mounted on the trucks and geared to the driving axles. Independent operation of either truck shall be provided.

3.12.6 Traction motors. Locomotive shall be equipped, with four series wound railway-type traction motors, ruggedly constructed, and ventilated by forced air, from independent blowers, or a single blower for each truck. One motor shall be mounted on each axle and shall drive the axle through suitable gearing.

3.12.7 Control. The locomotive shall have a single operating control station located on the right side of the operator's cab facing forward, and shall have multiple-unit equipment to enable the operation of two or more of same type locomotives from one operator's station.

3.12.7.1 Operating controls. The following controls shall be grouped and within easy reach of the operator when driving the locomotive in either direction:

(a) Master control handle with throttle and reversing lever with interlock to prevent reversal while power is being transmitted to the motors.
(b) Engine start pushbutton.
(c) Engine stop switch.
(d) Brake, sander, bellringer, horns, and window wiper valves.
(e) Emergency fuel shutoff.
(f) Heater switches.
(g) Light switches, inside and outside operator's cab.
(h) Headlight transfer switch.
(i) A foot operated safety (deadman) control switch.

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3.12.7.2 Instruments. All instruments shall be so mounted as to afford good visibility to the operator. Illumination of all gages for night operation shall be provided. Instruments shall include, but not be limited to the following:

(a) Brake gages.
(b) Battery charging ammeter.
(c) Engine lubricating oil pressure gage.
(d) Coolant water temperature gage.
(e) Fuel level indicator gage.

3.12.7.3 Pressure gage testing. All pressure gages shall be mounted with adapters for a gage tester to permit in-place testing.

3.12.7.4 Warning indicators. Automatic indicators shall be provided for the following conditions:

(a) Low lubricating oil pressure - audible with light.
(b) High coolant water temperature - audible with light.
(c) Tripped ground relay - audible with light.
(d) Sander operation - warning light.
(e) Wheel slip - audible with light.

3.12.7.5 Safety controls. Special safety controls shall include, but not be limited to the following:

(a) Engine overspeed trip (engine mounted) (see 3.12.1).
(b) Emergency engine shutdown (engine mounted) (see 3.12.4.1).
(c) Traction motor current limiting equipment (see 3.12.5 and 3.12.6).
(d) Combination speedometer and odometer (see 3.12.1).

3.12.7.6 Traction motor control. Traction motor control shall permit traction motors to make transition to series parallel with field shunting. A manually operated cutout switch mounted on cab panel shall be provided for isolating the motors on the front or rear truck.

3.12.8 Underframe. The locomotive underframe shall be of built-up welded steel or one piece cast steel construction, gusseted and reinforced with steel bumpers. The underframe shall be designed to resist and absorb all stresses due to single or multiple unit operation of four locomotives. Hooks shall be provided on the underframe side sills or truck frames for carrying two rerailers.

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3.12.8.1 Jacking pads, lifting lugs, and tiedown points. A minimum of four combination jacking pads and lifting lugs shall be provided on the underframe at the end and on the centerline of the bolster, with the location clearly indicated by stenciled markings of "SLING HERE". Safety hooks shall be provided to prevent slewing and to permit the trucks to be lifted with the superstructure. Two tiedown eyes with at least 1-1/4-inch (3.18 cm) holes shall be located on each end sill of locomotive frame and six padeyes sufficient to take at least a 7/8-inch (2.2 cm) shackle shall be welded to the under surface of each side sill for use as tiedown points. Tiedown points shall be clearly indicated by stenciled marking.

3.12.8.2 Lifting eyes. Lifting eyes for convenient lifting shall be provided on diesel engine, main generator. Auxiliary generator, excitor, air compressor, and traction motors.

3.12.9 Draft arrangements. Locomotives shall be designed for the type coupler arrangement specified (see 1.2). Provisions shall be made in the locomotive design for the AAR type coupler, Alliance or Sharon coupler, or Willison type coupler, drawhook with screw couplings, and buffers to be applied front and rear as specified (see 1.2). When side buffers are specified, they shall have a minimum capacity of 8,000 foot-pounds (10849 pa) end arrangement shall be as specified.

3.12.9.1 Automatic couplers. Couplers shall be of the configuration and height above the top of the rail as specified (see 1.2). Uncoupling mechanism, if applicable, shall be as specified. Top operation is preferred.

3.12.9.2 Draft gear. Draft gear shall be either rubber or friction type suitable for AAR draft gear pocket. AAR approved draft gear and yoke shall be used with AAR Type "E" couplers or Willison couplers.

3.12.9.3 Drawhook and screw couplers. Drawhook and screw couplers shall be UIC standard and as shown on applicable drawings. When used, each coupler shall consist of a drawhook with a link and turnscrew mechanism. Screw and link mechanism shall be supported by a bracket as shown on applicable end arrangement drawing.

3.12.9.4 Drawgear. The drawgear shall be friction, rubber, or spring type as shown on the end arrangement drawings furnished by the contracting officer or in accordance with UIC Code 520-OR and shall be compatible with drawhook and buffer arrangement.

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3.12.9.5 Side buffers. Side buffers shall be high capacity energy absorbing type in accordance with UIC Code 526-OR. Location shall be as shown on end arrangement drawings furnished by the contracting officer.

3.12.10 Trucks. The running gear shall consist of two four-wheel swiveled, equalized, pedestal-type motor trucks. Pedestal jaws shall be fitted with hardened spring steel liners. Wheels, axles, and motor shall be removable as a unit. Wheel guards shall be mounted in front of leading wheels at each end of locomotive and above centerline of the rail. Each truck shall have one-piece steel side frames with integral pedestals and cast steel centerplates. Centerplates shall be oil-lubricated, incorporating oil reservoirs and fitted with dust guards. Side bearings shall be in accordance with AAR requirements, having truck safety locks capable of permitting safe lifting of the locomotive with trucks attached during shipment, or rerailing of locomotive. Swiveling limit devices shall be provided.

3.12.10.1 Axles. Axles shall be furnished with journals suitable for application of roller bearings. Axle material shall conform to AAR requirements.

3.12.10.2 Wheels. Wheels shall be solid, wrought steel, multiple-wear, 40-inch (101.6 cm) diameter, with heat-treated rims 2-1/2 inches (6.4 cm) thick, and AAR standard contour of tread and flange. Wheels shall conform to AAR requirements. Wheels shall be pressed on the axles in accordance with AAR Wheel and Axle Manual.

3.12.10.3 Roller bearings. Roller bearings shall be lubricated with grease conforming to AAR requirements. Roller bearings shall be AAR approved design for 6-1/2 by 12-inch (16.5 by 30.5 cm, or equivalent) journals.

3.12.10.4 Springs. Long travel coil springs shall be used in the suspension system to assure full vertical travel of wheel and axle assemblies. Material shall conform to AAR.

3.12.11 Brake equipment. The locomotive shall be equipped with air or vacuum brakes as specified. Installation of all brake equipment, piping, and rigging shall not interfere with accessibility of the power plant and transmission and shall be in accordance with requirements of AAR, DOT and UIC Code 548-OR and recommendations of the manufacturer. Pipe and fittings shall conform to AAR requirements. Air hoses and couplings conforming to AAR requirements shall be installed at each end of the locomotive. One 1-1/4-inch (3.2 cm) minimum emergency brake valve shall be provided in a position operable from left side of cab at fireman's position.

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3.12.11.1 Airbrake equipment. The locomotive shall be equipped with both independent and automatic airbrakes and shall incorporate a safety deadman with alert combination control feature. Airbrake equipment and installation shall conform to AAR requirements or Power Brake Laws.

3.12.11.2 Vacuum brakes. When specified, the locomotive shall be equipped with vacuum brakes conforming to UIC Code No. 548-OR unless otherwise specified herein. British Whitworth pipe thread series shall be furnished on vacuum brake components and on the ends of pipes connecting to components.

3.12.11.3 Dynamic braking. The locomotive shall have dynamic braking equipment, when specified (see 6.2).

3.12.11.4 Handbrake. An AAR approved type handbrake, located inside the cab and accessible to the operator, shall be provided.

3.12.11.5 Compressor. An air compressor of the water-cooled, two-stage type of required capacity for the specified brake system shall be provided. Compressor shall be mechanically driven by a power takeoff from engine. The compressor shall be provided with an alcohol device installed at the compressor's discharge.

3.12.12 Superstructure. The superstructure, consisting of the power plant enclosure and an operator's cab, shall be of fabricated steel construction and securely fastened to the underframe.

3.12.12.1 Power plant enclosure. The power plant enclosure shall be bolted to the underframe and readily removable for maintenance purposes without disturbance to the operating equipment of the locomotive. The enclosure shall be sectionalized over the engine, generator, and radiator compartment so that sections can be removed separately. Hatches with hinged covers shall provide complete accessibility to the engine, governor, oil filters, compressor, generator, and engine coolant expansion tank and thermostats. Hatch covers shall be adjusted to remain in open position and secured in the fully closed position on the inside of the enclosure. Louvered openings in the side doors shall be arranged for the installation of impingement type air filters. The radiator and fan openings shall be protected by heavy screen wire. The cooling compartment shall be provided with automatically operated shutters (see 3.8.2).

3.12.12.1.1 Ventilation. Adequate ventilation for engine, operator's cab, generator, traction motors, air compressor, and control apparatus shall be provided.

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3.12.12.2 Operator's cab. The operator's cab shall be on the front of the locomotive with the power plant enclosure trailing, and arranged to give the operator a clear view to the front and rear for operation in either direction. The cab shall be insulated against sound and heat. The cab shall be weathertight, with doors, window sills, and frames weatherstripped to prevent water leaking into the cab. The flooring shall be supported so that any equipment under the floor is easily accessible.

3.12.12.2.1 Insulation. Cab sides and roof shall each be insulated with fiber glass or polyurethane insulation. Insulation shall be self-extinguishing when removed from an open flame. Cab with its insulation shall provide operator comfort at minus 25° F (minus 31.7° C).

3.12.12.2.2 Windows and doors. The cab shall have safety glass windows in the front, rear, and on each side to provide visibility in all directions. The center windows on each side shall have sliding sash equipped with latches. One door at the rear shall provide access to walkway along power plant enclosure and one door at the front shall provide access to the front platform. The doors shall be provided with keyed locks.

3.12.12.2.3 Equipment. One upholstered, swivel type seat, with back rest, adjustable for height and located to enable operation in either direction shall be located at the operator's station. Upholstered or moulded rubber armrests shall be provided. A metal, box-type seat for storing small tools, with a hinged, cushioned seat suitable for two men, and a backrest capable of being positioned for either direction of operation, shall be located at the left side of the cab. Four pneumatically operated window wipers shall be mounted on the cab windows for visibility in either direction of motion. Four sun visors shall be located inside the cab, one at front and rear vision window at both operator's and helper's stations. Provision shall be made in toolbox for storing flags, fusees, and torpedoes. An inspection card holder shall be located on the rear wall. Two 20-pound (9.1 kg) dry chemical powder type fire extinguishers shall be provided, one in the operator's cab and one in the power plant enclosure. Bay windows shall be installed on each side of the cab.

3.12.12.2.4 Cab heaters. The operating cab shall be heated by means of electric fan-type heaters connected with engine cooling system with shutoff valves. The heaters shall have controls located at operator's position. The heaters shall be piped into the engine cooling system for delivery at all times of an adequate flow of coolant at essentially the temperature of the engine coolant outlet. Heater locations shall permit convenient access to pipe connections and to the electric motor brushes and any other components requiring maintenance. Pipe connections at cab

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heaters shall be provided with steel, flexible, threadless type connections in place of rubber hose.

3.12.12.2.5 Defrost system. Heating facilities shall be provided to defrost and deice at least 75 percent of the front and rear cab window areas, under temperature requirements specified herein.

3.12.12.2.6 Walkways. Platform walkways shall be provided at each end of the locomotive and along the power plant enclosure with sidesteps located at each corner. A pilot, with side steps conforming to FRA requirements shall be attached to each end plate. Walkways and steps shall have handrails and nonskid treads.

3.12.13 Wiring and lights.

3.12.13.1 Wiring. Schematic diagrams and wiring diagrams shall be furnished. All wiring shall conform to AAR requirements. All connections shall be soldered or joined by clasp and crimp type solderless connectors. Wiring connections, except motor leads, shall be made only at junction boxes, board, terminal strips, and at the apparatus to be operated. Wires shall not be spliced. All lighting and control circuits shall be protected by circuit breakers readily accessible. All electrical and mechanical interlocks shall be enclosed in dustproof coverings. The locomotive frame, body, and similar components shall not be used as a ground. All wiring shall be moisture-resistant, neoprene-jacketed, or equal, and all control wiring shall be placed in conduit. The wiring diagrams shall fully indicate and identify all electrical circuits and equipment, to permit tracing and repair of all circuits. Tagging and identification of all conductors shall be with nonmetallic tape applied near each end of the conductors.

3.12.13.2 Lights. The locomotive shall have twin-sealed beam headlights, front and rear, and class lights, in accordance with DOT requirements. The locomotive shall have four marker lights, two on each side, front and rear, and shall have number box lights on the front and rear ends. The locomotive shall have lights at each side step and near the end of each walkway, a track light, lights for illuminating the power plant enclosure and control compartment, a cab dome light and instrument lights. An extension light, with a 30-foot (9 m) minimum length of cord, and a receptacle for the same, shall be located in the power plant enclosure.

3.12.14 Sanding system. Outboard and inboard sanders shall be furnished and arranged to sand ahead of the lead wheels in each direction of travel. Sanders shall operate automatically when emergency brake applications are made.

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3.13 Lubrication. Unless otherwise specified, all surfaces that require lubrication shall be properly lubricated with grease according to the component manufacturer's recommendations.

3.14 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below shall be protected from the effects of fungus growth and moisture by an overall treatment with a waterproofing varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.15 Painting and marking.

3.15.1 Painting. Painting shall conform to MIL-P-3321 with the following exceptions:

(a) Surface preparation: Surface cleaning shall not be required for the car body constructed of blased, pickled, or phosphate treated material before primer coating.
(b) Priming: The exterior of the carbody shall be primed with one coat of an alkyd resin type primer 0.75-1.25 mils D.F.T. For intermediate primer surfacer, one coat of an acrylic primer surfacer shall be applied to the carbody exterior to a 0.5 to 1.0 mil D.F.T.
(c) Finishing: The interior of the carbody and all exposed surfaces of the underframe are painted with an alkyd enamel 1.0 to 1.5 mils D.F.T.
(d) Lacquer system: The exterior of the carbody shall be painted with three coats of acrylic lacquer of the appropriate colors to a minimum of 1.5 mil D.F.T. per color.

3.15.2 Marking. Marking shall conform to MIL-P-3321, as specified.

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3.15.3 Piping and fittings. Piping and fittings shall be painted to indicate the respective fluids carried by the piping. Color code of the marking shall be as follows:

Lubricating oil - yellow
Fuel oil - red
Cooling water - blue
Compressed air - green
Vacuum - gray
Exhaust gas - white when applicable

3.16 Identification plates.

3.16.1 Components. Each of the major component parts of the locomotive shall be provided with an identification plate bearing the name of the manufacturer, manufacturer's serial number, and all other pertinent information usually furnished by the manufacturer.

3.16.2 Locomotive. An identification plate in accordance with MIL-P-3558, Type I, shall be permanently attached to the frame of the locomotive in a visible location. The following information shall be stamped or embossed thereon:

(a) National Stock Number (NSN).
(b) Item description (nomenclature).
(c) Specification number.
(d) Contract number.
(e) Name of builder.
(f) Builder lot number.
(g) Date of manufacture (month and year).

3.17 Accessories.

3.17.1 Horns. The locomotive shall be furnished with two, single-tone horns, one facing forward and one facing to the rear.

3.17.2 Rerailers. The locomotive shall be furnished with two rerailers. One inside and one outside for size of rail as specified (see 6.2).

3.17.3 Turning bar. The locomotive shall be furnished with an engine turning bar.

3.17.4 Grease gun. If the locomotive has grease fittings for lubrication, one 15-ounce, lever type grease gun, complete with the necessary fittings shall be furnished.

3.17.5 Bell. The locomotive shall be furnished with one nonswinging type bell with air operated ringer.

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3.17.6 Engine exhaust muffler. The locomotive shall be furnished with one muffler per engine.

3.18 Noise limits. The noise level at the operator's station shall not exceed the limits specified in MIL-STD-1474, Catagory C for steady state noise when tested in accordance with 4.5.2.2.3 (aural nondetectability and impluse noise do not apply). The precautions of MIL-STD-1474 shall be provided if the sound level exceeds 85dB(A) when tested in accordance with 4.5.2.2.3.

3.19 Workmanship. The finished locomotive shall be free from any defects that could cause malfunctioning of components or in any way jeopardize or interfere with its safe operation. Construction of the locomotive and appearance of the applied coatings shall be compatible with standard quality workmanship for this type of equipment and in accordance with AAR requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspections as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with MIL-L-58009 or applicable AAR specifications as specified herein or that the components or assemblies have been previously approved by AAR.

4.1.2 Acceptability criteria. Locomotives which conform to all requirements in Sections 3 and 5 of this specification and pass all applicable examinations and tests in Section 4 of this specification will be considered acceptable by the Government.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced locomotive(s) inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

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4.3 First-produced locomotive(s) inspection. The first-produced locomotive which shall have been produced from production tooling shall be examined and tested to determine conformance to the requirements of this specification.

4.3.1 Examination. The first-produced locomotive(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced locomotive(s) shall be tested as specified in 4.5.2 through 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each locomotive shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each locomotive shall be subjected to a running test as specified in MIL-L-58009. Nonconformance to MIL-L-58009 and this specification shall constitute failure.

4.5 Inspection procedures.

4.5.1 Examination. Each locomotive shall be examined for the following defects:

101. Material not as specified (see 3.3).
102. Components and parts not interchangeable with similar components and parts (see 3.5).
103. Threads not as specified (see 3.4).
104. Safety provisions not as specified (see 3.6).
105. Parts subject to inspection, adjustment, or replacement not accessible (see 3.7).
106. Dimensions and weights not as specified (see 3.10).
107. Fuel system not as specified (see 3.12.4).
108. Standby battery-charging system not as specified (see 3.12.3.1).
109. Battery not as specified (see 3.12.3).
110. Battery-charging cables not as specified (see 3.12.3.2).
111. Traction motors not as specified (see 3.12.6).
112. Control equipment instruments, gages, and warning system not as specified (see 3.12.7).
113. Jacking pads and lifting lugs not as specified (see 3.12.8.1).
114. Tiedown points not as specified (see 3.12.8.1).
115. Lifting eyes not as specified (see 3.12.8.2).
116. Couplers not as specified (see 3.12.9).
117. Axles not as specified (see 3.12.10.1).

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118. Wheels not as specified (see 3.12.10.2).
119. Brakes not designed, arranged, or operated as specified (see 3.12.11).
120. Handbrake not as specified (see 3.12.11.4).
121. Air compressor not as specified (see 3.12.11.5).
122. Superstructure not constructed and equipped as specified (see 3.12.12).
123. Wiring and lights not as specified (see 3.12.13).
124. Engine coolant heater, when required, not as specified (see 3.8).
125. Sanding system not as specified (see 3.12.14).
126. Pipe and fittings not as specified (see 3.12.11 and 3.15.3).
127. Painting and marking not as specified (see 3.15).
128. Identification plates missing or not as specified (see 3.16).
129. Accessories to be furnished (see 3.17).
130. Workmanship not as specified (see 3.19).
131. Parts or components not as specified (see Section 3).
132. Parts or components missing (see Section 3).
133. Assembly incomplete or incorrect (see Section 3).

4.5.2 Tests. Each locomotive shall be tested in accordance with MIL-L-58009. In addition to the temperature requirements (see 4.5.2.1) and noise level measurements (see 4.5.2.2). Nonconformance to MIL-L-58009 and this specification shall constitute failure.

4.5.2.1 Temperature requirements. Engine tests at extreme ambient temperatures specified shall be conducted in accordance with MIL-STD-1400.

4.5.2.1.1 High temperature test. The high temperature test shall be conducted in accordance with Test Method 1440, MIL-STD-1400. Nonconformance to Test Method 1440 shall constitute failure.

4.5.2.1.2 Low temperature test. The low temperature test shall be conducted in accordance with Test Method 1450, MIL-STD-1400. Nonconformance to Test Method 1450 shall constitute failure.

4.5.2.2 Noise level measurement. Measure the noise at the operator's station in accordance with MIL-STD-1474 for steady state noise while operating the locomotive. Nonconformance to 3.18 shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Examination. Each locomotive shall be examined for the following defects:

134. Locomotive not prepared for delivery as specified.
135. Damaged paint not repaired as specified.
136. Exposed machined surfaces not preserved as specified.
137. Matchmarking of components not as specified.
138. Locomotive not cleaned as specified.
139. Depreservation guides not furnished and located as specified.
140. Markings not as specified.
141. Transporting railway car not marked as specified.

5. PACKAGING (see 6.4).

5.1 Lift configuration. The locomotive shall be preserved and packed as a single unit capable of being hoisted in a single lifting operation.

5.2 Removed assemblies and components. Bolts, nuts, pins, washers, and similar components shall be placed in one of the mating parts and secured to prevent loss. Small components, such as hardware and hoses, removed from their installed or operating location shall be consolidated in close-fitting wood boxes. Intermediate packaged components shall be blocked, braced, and cushioned within the container to prevent damage to the contents or container.

5.3 Matchmarking. Removed assemblies and components shall be matchmarked with lusterless, white enamel letters to facilitate reassembly. Matchmarking shall be overcoated with varnish. Components too small to accomodate stencil markings shall be identified with metal tags if available, or with waterproofed cloth shipping tags, attached to the mating parts.

5.4 Repainting. Surface of the locomotive or equipment on which paint has been damaged shall be cleaned and touch-up painted with paint matching the original paint finish.

5.5 Locomotive cleaning. The locomotive shall be clean of oil or grease drippings and smudges. The locomotive shall be free of all foreign material.

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5.6 Preservation. Preservation of the locomotive, which includes, but is not limited to, engine and auxiliary equipment, generator, traction motors, brake cylinders, piston rods, air and vacuum lines and equipment, roller bearings, technical publications, shall be in accordance with standard commercial practice for overseas shipment as recommended by the contractor, subject to review, modifications to the recommendations, and final approval by the contracting officer (see 5.10).

5.6.1 Exposed machine surfaces. In addition to the preservation specified in 5.6, unpainted exposed machine surfaces shall be coated with a preservative compound. When a soft nondrying compound is used, the coated surface shall be wrapped or covered with greaseproof, waterproof paper.

5.6.2 Depreservation Guide. Two copies of completed Depreservation Guide conforming to DA Form 3257 - Preservation and Depreservation Guide for Railroad Equipment, shall be furnished with each locomotive. Each copy shall be placed in a waterproof envelope and marked with the words "Depreservation Guide". One copy of the Depreservation Guide shall be attached to each side of the locomotive in a conspicuous place.

5.7 Packing. Packing of the locomotive shall be in accordance with standard commercial practice for overseas shipment as recommended by the contractor subject to review, modifications to the recommendations, and final approval of the contracting officer (see 5.10).

5.8 Carloading of the locomotive. When specified, the locomotive shall be loaded on a transporting railway car in accordance with AAR rules or other applicable rules. Markings of the transporting railway car shall be as specified in 5.9.2. Domestic service locomotives will be shipped on their own wheels. Foreign service locomotives may also be shipped on their own wheels if proper couplers are available.

5.9 Marking. In addition to special marking specified in the contract or order, marking for shipment shall be in accordance with MIL-STD-129 and as specified herein.

5.9.1 Warning markings. Appropriate warning markings shall be conspicuously displayed when moving or operation of the locomotive may be injurious to the locomotive prior to depreservation.

5.9.2 Placard for transporting railway cars. When locomotives are loaded on a transporting railway car, AAR approved placards with the words "DO NOT HUMP" printed thereon, or other applicable and similar cards, shall be displayed on the transporting railway car in accordance with the applicable rules governing the use of such cards.

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5.10 Submittal of preservation and packing plans. The contractor shall submit preservation and packing plans to the contracting officer at the time specified (see 6.2).

6. NOTES

6.1 Intended use. The locomotives covered by this specification are intended to be used for railway switching and road service in either single or multiple-unit operation on domestic or foreign railways.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Gage of trucks to be furnished (see 1.2).
(c) Type of brake equipment to be furnished (see 1.2).
(d) Type of couplers to be furnished (see 1.2 and 3.12.9).
(e) Time frame required for submission of first-produced model(s) (see 3.2).
(g) Whether AAR structural analysis is required (see 3.12).
(h) When battery is to be shipped dry (see 3.12.3).
(i) When side buffers are to be furnished (see 3.12.9.5).
(j) Size of rerailers to be furnished (see 3.17.2).
(k) When locomotive shall be loaded on a transporting railway car (see 5.8).
(l) Time for submittal of preservation and packing plans to the contracting officer (see 5.10).

6.3 Configuration. Prior to use of this specification for procurement, the following data for the applicable railway shall be verified by the contracting officer:

(a) Conformance of equipment to clearance diagram AAR Plate C or MS35833.
(b) Gage.
(c) Brake system.
(d) Coupler arrangement.
(e) Wheel flange thickness and tread contour.

Upon receipt of the above information, one of the below listed drawings as applicable (see 2.1), may be used for procurement:

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ME

13216E7927 - Broad Gage Fleet; End Arrangement of Coupler, AAR Type.
13216E7928 - Broad Gage Fleet; End Arrangement of Coupler, Drawhook and Screw Coupling.
13216E7929 - Broad Gage Fleet; End Arrangement of Coupler, Willison Type.

6.4 Packaging. The requirements of Section 5 are the minimum acceptable for the subject locomotives. Since some locomotives may be destined for foreign service, it is not likely that such FS Locomotives will be held in domestic storage; however, it is anticipated that the shipment overseas will involve deck loading and a storage period time not to exceed 30 days.

6.5 First-produced locomotive. Any changes or deviations of production locomotives from the approved first-produced locomotive during production will be subject to the approval of the contracting officer. Approval of the first-produced locomotive will not relieve the contractor of his obligation to furnish locomotives conforming to this specification.

6.6 DA Form 3257. The contracting officer should arrange to furnish DA Form 3257 when requested by the contractor.

6.7 Cab keys. The contracting officer should arrange for forwarding the cab keys to the receiving activity at time of shipment.

6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each locomotive.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary including any special tools, and instructions on shipment of locomotives. A suggested paragraph is as follows:

"Shipment of locomotives shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.10 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodian:
Army - ME

Preparing activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-L-52731A(ME) Locomotives, Diesel-Electric, 56-1/2-, 60-, 63-, or 66-In. Ga., 100-Ton, 0-4-4-0 Whl f/Domestic/Foreign Svc

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